



Machine Learning

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Content (Week)

- Machine Learning
- Its types
- Supervised learning
- Supervised learning algorithms
- Implementation of algorithms

Content

- Machine Learning
- Its types
- Regression problems
- Classification problems

What is Machine Learning?

- One of the criticism of AI was, a machine is not intelligent if it can not learn.
- Learning means doing the similar task efficiently next time.
- **Machine learning is the science of getting computers to act without being explicitly programmed.**
- Learning programs are adaptive.
- requires data.

Why?

- Learning vs Memorization
- Lots of data available.
- Can handle voluminous, complex data to give accurate results.
- To identify pattern from data.

Examples

- Friend suggestion on Social Media platforms.
- Product/Video recommendation.
- Automatic spam filtering

Types of learning

- Supervised
- Unsupervised
- Reinforced
- Semi-supervised

Supervised learning

- learns from example data and associated outputs are already given.
- predicts the correct response for unseen examples.
- similar to human learning under the supervision of a teacher.
- Ex: Classroom Teaching

Supervised learning

Year	Manufacturing Cost (in million dollar)	Advertising Cost (in million dollar)	Sales (in million dollar)
1991	27	1	35
1992	30	1.67	35.23
-	-	-	-
2020	50	5.66	59.23

Year	Manufacturing Cost (in million dollar)	Advertising Cost (in million dollar)	Sales (in million dollar)
2024	66	8.9	?

Supervised learning

Email	Spam/Not spam
Dear Lucky Winner, You have been randomly selected in our Big Money Giveaway! Click the link below to claim your prize of \$1,000,000 immediately.	Spam
Hi Team, This is a reminder for our weekly project sync-up meeting scheduled for tomorrow at 3 PM in the main conference room.	Not Spam
- - - - -	-----
- - - - -	-----

Email	Spam/Not spam
Hello, Are you suffering from any chronic pain! Our new miracle drug cures all diseases. Order now and receive a 50% discount on your first purchase.	?

How Supervised Learning Works?

- The training process.
- Different algorithms use different approaches for learning.
- Major tasks: Regression and classification
- Major algorithms:
 - Linear Regression
 - Logistic Regression
 - Decision Tree
 - Artificial Neural Network (ANN)

Regression

- A supervised learning technique.
- When the output has real values (continuous variable).

x1	x2	x3	Output
2	3	1	2.3
4	5	2	3.1
6	7	5	7.8
-	-	-	-
-	-	-	-
3	5	4	?

Regression

- Example:
 - What is the sale given advertising cost and manufacturing cost?

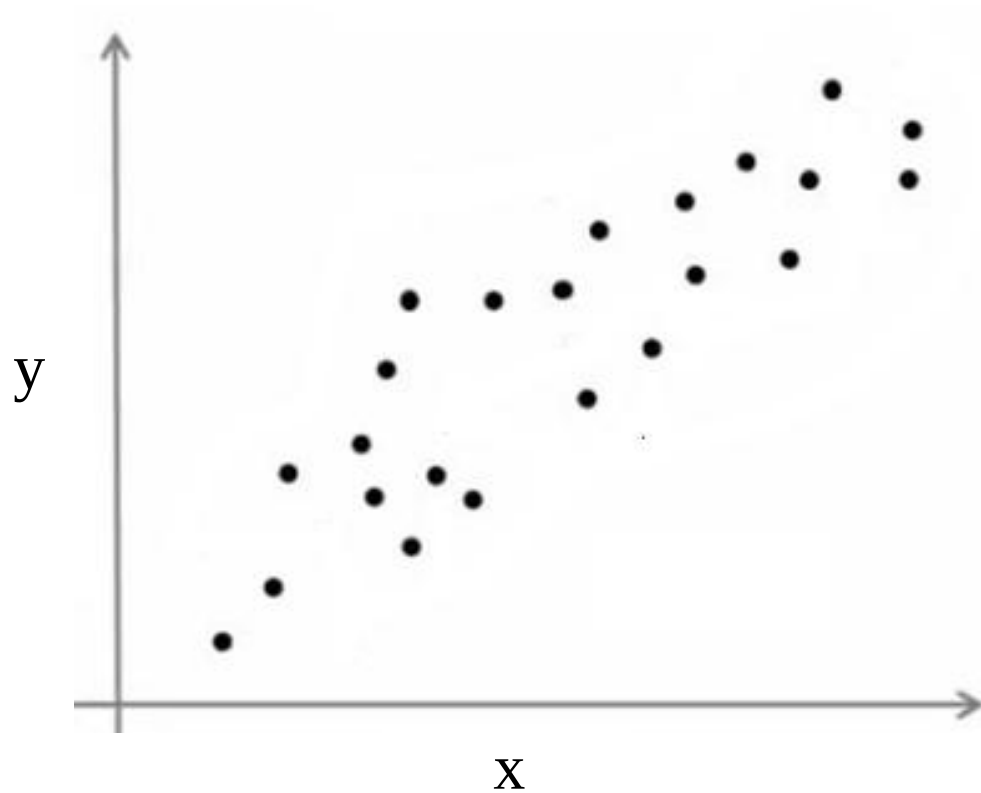
Year	Manufacturing Cost	Advertising Cost	Sales
1991	27	1	35
1992	30	1.67	35.23
-	-	-	-
-	-	-	-
-	-	-	-
2020	50	5.66	?

Regression

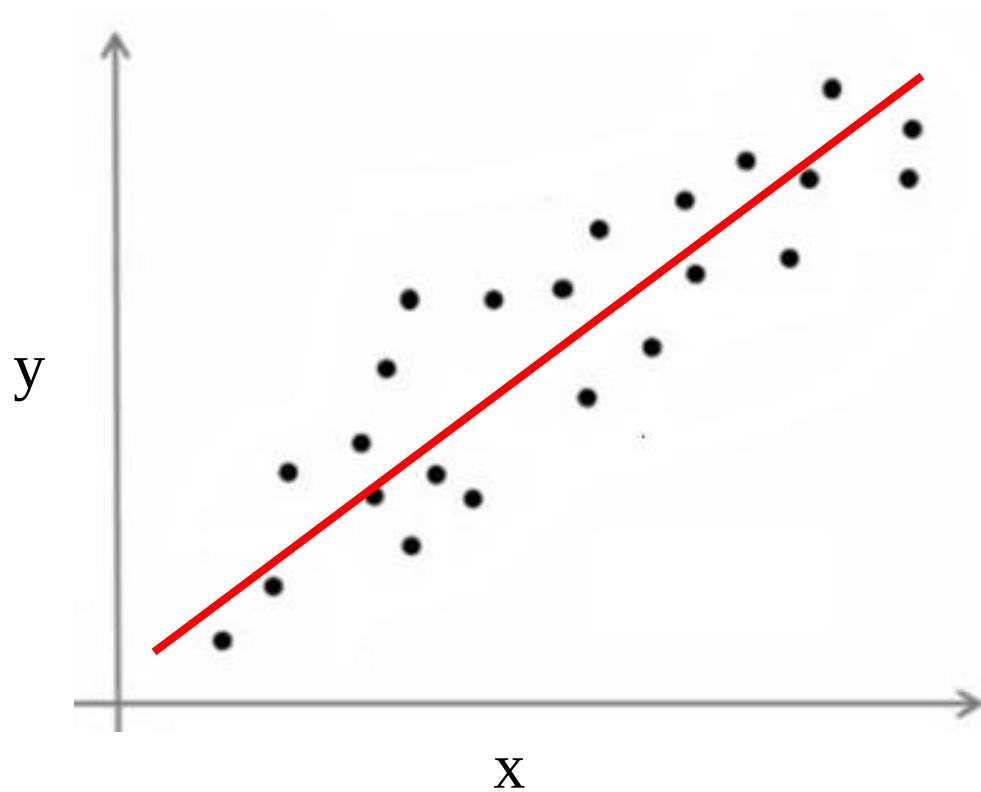
- Example:
 - How much rain will fall in July this year in Mumbai?

Year	Humidity	Temperature	Rain
2010	67	38	10
2011	75	35	12.34
-	-	-	-
-	-	-	-
-	-	-	-
2019	70	38	?

Example



Example



Types

- Linear Regression

- **Assumption:** Linear relationship between independent variable (x) and dependent variable (y).

$$y = \beta_0 + \beta_1 x + \epsilon$$

- Polynomial Regression

- **Assumption:** Non-linear relationship between independent variable (x) and dependent variable (y).

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n + \epsilon$$

- Multiple Regression

- two or more independent variables and a single dependent variable.

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

Classification

- A supervised learning technique.
- When the output is any one from fixed set of values (discrete variable).

x1	x2	x3	Output
2	3	1	class1
4	5	2	class2
6	7	5	class1
-	-	-	-
-	-	-	-
3	5	4	?

Classification

- Example: Whether game can be played tomorrow?

Humidity	Wind speed	Temperature	Game Possible?
70	34	27	Yes
64	50	35	No
67	43	33	Yes
-	-	-	-
-	-	-	-
66	39	22	?

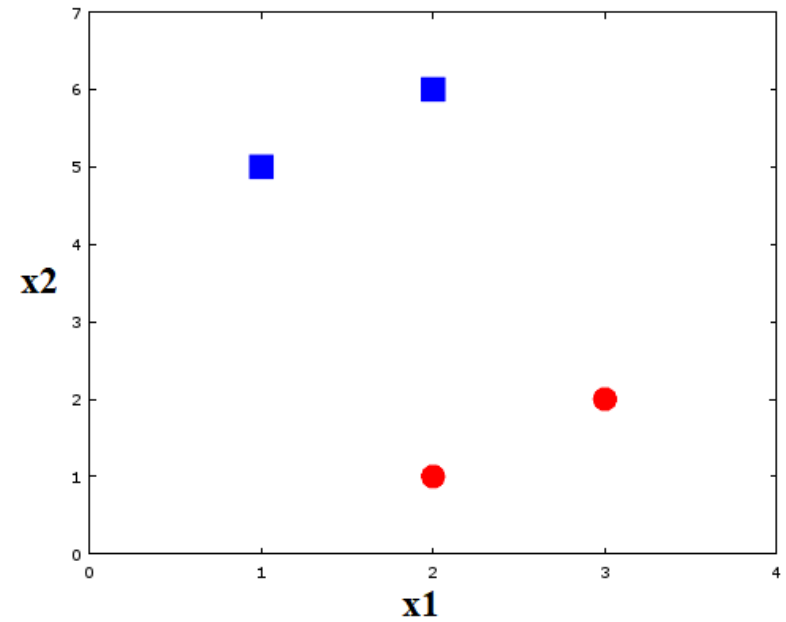
Classification

- Example: What is the level of cancer?

Age	Gender	Mediastinum	Cancer
55	Female	Yes	Level3
65	Male	No	Level1
60	Male	Yes	Level2
-	-	-	-
-	-	-	-
71	Female	No	?

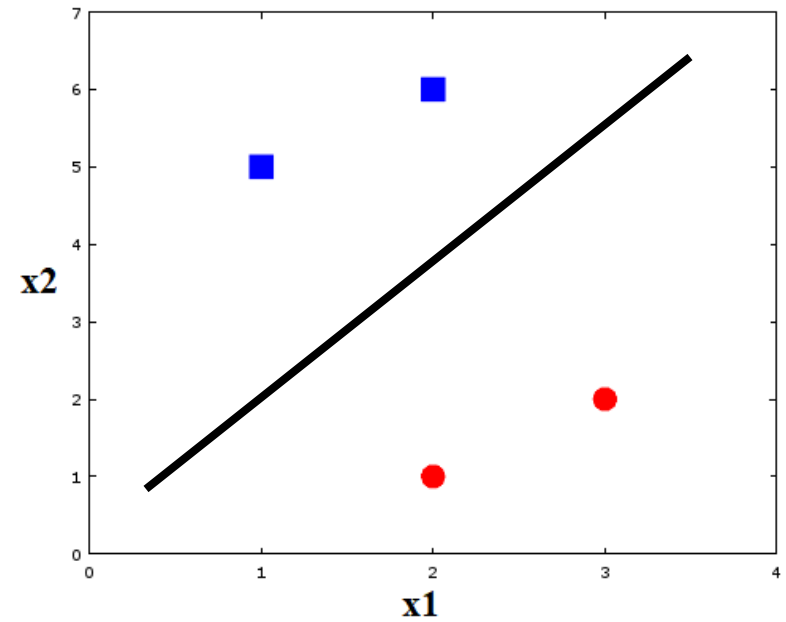
Example

x1	x2	y
2	1	Yes
3	2	Yes
1	5	No
2	6	No



Example

x1	x2	y
2	1	Yes
3	2	Yes
1	5	No
2	6	No



Summary

- Machine Learning
- Supervised machine learning
 - Regression
 - Classification
- Unsupervised machine learning
- Reinforcement learning
- Semi-supervised